Evaluating Anomaly Detectors on Energy Time Series: What Drives the Rankings?

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Abstract

Choosing an anomaly detector for energy data is hard because general benchmarks say that no method wins everywhere, but not why rankings change or how much each design choice matters. We study this on energy demand and price series with documented events (GB and Texas demand, commodity prices, a simulated European grid), under one written evaluation protocol. First, a previously proposed rule linking event duration and sampling resolution to the best detector does not hold. It came from mixing event-level and timestamp-level metrics, an uncontrolled alert budget and confounded variables, and a held-out test on 862 labelled public series does not recover it (partial Spearman $\rho = -0.05$, 95% interval $[-0.16, 0.14]$). Second, a fully crossed semi-synthetic design (648 datasets, 9 detectors, 2 feature sets) shows that the detector explains 19.9% of the variance in detector skill, while detector × setting interactions explain about 26%, so rankings depend on event type, duration and magnitude. Third, on real GB and ERCOT data with equal-budget tuning and paired block-bootstrap intervals, simple baselines (a seasonal-naive residual, a matrix profile) match or beat LSTM and Transformer autoencoders, the winner depends on the event, and two events per series cannot rank detectors. We also test whether the event labels themselves are valid, and release code, tests and all result files.

1 Introduction

Choosing an anomaly detector for energy time series is a practical problem, and the general benchmarks (TSB-UAD [5], TSB-AD [7] and TimeEval [3, 4]) already report that no detector wins everywhere and that simple methods are often competitive. The open question for an energy analyst is more specific: given *this* kind of event on *this* kind of series, which design choices—detector family, input features, event type and duration, sampling resolution, evaluation metric—actually move the result, and by how much? Answering it requires holding the pipeline fixed while one thing changes, and real energy series with documented events (one or two per series) cannot do that alone.

The work began from a five-detector comparison on four datasets that proposed a *duration–resolution rule*: reconstruction-based detectors (an LSTM or Transformer autoencoder) should gain over density-based ones (DBSCAN, k-NN) as an event spans more native samples, and DBSCAN's F1 should fall as the sample count grows. Re-examining that evidence showed that it mixed metrics of different kinds, confounded the variables (series, resolution, prevalence and duration changed together), rested on one or two events per dataset, tuned detectors unequally, left the protocol under-specified and did not examine label validity. This paper reports what the evidence supports once these problems are removed, and how that connects to the rule: the rule is not supported when metric, alert budget and prevalence are held fixed (Sec. 6) or on held-out series (Sec. 9); in its place, detector rankings turn out to be conditional on event type, duration, magnitude and input features (Sec. 7); and on real GB and ERCOT data, simple baselines are competitive with deep autoencoders while two events per series cannot rank detectors (Sec. 8). We propose no new detector. The contributions are empirical and methodological:

1. **A specified protocol** (Sec. 5): exactly what the models see, how dated events become labels, how alerts are formed and how each metric is computed, with event-level and timestamp-level metrics never mixed.
2. **Direct tests of label validity** (Sec. 4): whether the series deviates inside labelled events, how many unlabelled days deviate as much, and whether detector rankings survive plausible label errors.
3. **Attribution, not just ranking**: a fully crossed semi-synthetic design (Sec. 7) that separates event type, duration, magnitude, resolution, background, feature set and detector, and a held-out test of the earlier rule on 862 public series (Sec. 9).
4. **A real-data comparison with stronger baselines and uncertainty** (Sec. 8): equal-budget tuning, simple and distance-based baselines from the benchmark literature, paired block-bootstrap intervals, and per-event coverage.
5. Released code with tests, including regression tests for the evaluation conventions (inclusive event end day, window seams, VUS-PR resampling).

Organisation. Section 2 reviews related work; Section 3 describes the series, the inputs the detectors see and how events become labels; Section 4 tests label provenance and validity; Section 5 specifies the evaluation protocol (pipeline, detectors, metrics and uncertainty); Section 6 gives the controlled duration–resolution sweep; Section 7 the fully crossed attribution design; Section 8 the real-data comparison; Section 9 the held-out test of the duration–resolution rule

on public series. Sections 10–12 give a checklist, the limitations and the conclusion.

2 Related Work

Several benchmark and evaluation studies are directly relevant. TSB-UAD [5] collects 13,766 labelled univariate series and evaluates twelve detectors; TSB-AD [7] curates 1,070 series from 40 datasets, benchmarks statistical, neural and foundation-model detectors (33 are listed in its repository), recommends VUS-PR as the most reliable measure and reports that simpler methods often outperform advanced neural networks. Schmidl et al. [3] re-implemented 71 detectors and evaluated them on 976 series, finding no single best algorithm and that deep methods were not competitive with simpler ones; their toolkit is [4]. Wu and Keogh [2] document triviality, unrealistic anomaly density, mislabelled ground truth and run-to-failure bias in common benchmarks. Kim et al. [1] show that point-adjusted F1 can make a random score look state-of-the-art, and Paparrizos et al. [6] argue for threshold-independent range-aware measures. Evaluation on events rather than timestamps has its own literature: Lavin and Ahmad's NAB [9] scores detections inside windows around labelled anomalies, Hundman et al. [10] count a detection as correct when it overlaps a labelled anomaly sequence, and Tatbul et al. [11] generalise precision and recall to ranges. Our event-level and timestamp-level metrics (Sec. 5) follow these conventions and are never combined. The survey of Darban et al. [8] and its companion list of deep methods, datasets and evaluation metrics (`https://github.com/zamanzadeh/ts-anomaly-benchmark`) catalogue about fifty benchmark datasets and ten evaluation metrics and give guidance on when to use each.

Positioning. These works establish, across hundreds to thousands of generic series, that rankings are dataset-dependent and that simple baselines are strong. We therefore do *not* claim "no universal detector" as a finding. What they do not provide, and what this paper adds for energy data, is (i) a *decomposition* of how much of the variation in detector skill is due to the detector, the event type, its duration and magnitude, the sampling resolution and the input features, measured on a fully crossed design where those factors are independent; (ii) a protocol written down at the level needed to reproduce an energy study with dated external events; (iii) direct tests of whether such event labels are valid; and (iv) an account of how a plausible-looking rule emerged from confounded variables and failed a held-out test. Energy series differ from the generic benchmarks in having strong multi-scale seasonality and labels that are *documented external events* (storms, a pandemic) rather than annotated points, which is why the label-validity question is not secondary here.

3 Data, Inputs and Events

Series and what the detectors see. Table 1 lists every series and event with its native-sample count, and Table 2 gives the structure of each dataset. All detectors in this paper are *univariate*: each series is analysed on its own, and the model input is a handful of features derived from that one series (Sec. 5). No detector uses relations between series. This is a deliberate scope limit, stated here because the four study datasets comprise six series with different structure:

- **WTI and Henry Hub** (FRED, daily): two commodity price series sharing the same two documented events; they are not independent replicates of one another.
- **Simulated French grid** (Gillioz et al. [14]): 4,097 buses are aggregated to one national load series, and 62 lines to 28 border crossings, of which the line-flow series is one. These are *simulated*; the "COVID-19" and "nuclear crisis" events are imposed or attenuated by the simulation, so this study tests detectors against a known generating mechanism rather than against observed history.
- **GB national demand** (NESO, half-hourly, observed) and **ERCOT demand** (EIA-930, hourly, observed): single aggregate series with two events each.

The 11 event rows in Table 1 contain 8 distinct labelled windows: three rows repeat a window already counted for another series (Henry Hub repeats both WTI events; French line flow repeats the French load nuclear-crisis window).

Events as labels. An event is a pair of inclusive calendar dates; every timestamp from 00:00 of the start day to 24:00 of the end day is labelled anomalous. The end date is inclusive. An early release of the evaluation code treated it as exclusive and so dropped the last day of every event (for a 5-day event, one fifth of its labelled timestamps); the behaviour is fixed, covered by a regression test, and documented in `docs/KNOWN_ISSUES.md` of the repository [22]. Events were specified from external sources *before* any detector was run; they are not derived from the series.

4 Label Provenance and Validity

A detector that flags an unlabelled period cannot show that the period is a true event, so a label set cannot be validated by reporting that detectors "find" events the labels miss. We therefore test validity without using detectors where possible.

Provenance and tiers. Tier 1 events are documented crises fixed before modelling (Texas Freeze, Winter Storm Elliott, the 2020 lockdown, the December 2022 cold snap, the two market events). Tier 2

Series	Resolution	Data	Period	Event	Dates (inclusive)	Days	Native samples
WTI spot (FRED)	daily	real	2018–2023	Texas Freeze	2021-02-10 – 2021-02-25	16	≈11 (business days)
WTI spot (FRED)	daily	real	2018–2023	Russia–Ukraine	2022-02-20 – 2022-06-30	131	≈94 (business days)
Henry Hub (FRED)	daily	real	2019–2023	Texas Freeze	2021-02-10 – 2021-02-25	16	≈11 (business days)
Henry Hub (FRED)	daily	real	2019–2023	Russia–Ukraine	2022-02-20 – 2022-06-30	131	≈94 (business days)
FR load (Gillioz)	hourly	synthetic	2016–2020	Nuclear crisis	2016-10-10 – 2017-01-27	110	2,640
FR load (Gillioz)	hourly	synthetic	2016–2020	COVID-19	2020-03-16 – 2020-05-11	57	1,368
FR line flow (Gillioz)	hourly	synthetic	2016–2018	Nuclear crisis	2016-10-10 – 2017-01-27	110	2,640
GB demand (NESO)	half-hourly	real	2016–2023	COVID-19	2020-03-23 – 2020-05-31	70	3,360
GB demand (NESO)	half-hourly	real	2016–2023	Cold snap	2022-12-08 – 2022-12-17	10	480
ERCOT demand (EIA-930)	hourly	real	2019–2023	Texas Freeze	2021-02-14 – 2021-02-19	6	144
ERCOT demand (EIA-930)	hourly	real	2019–2023	Elliott	2022-12-22 – 2022-12-26	5	120

Table 1: Series, resolution, data type and events with duration in calendar days and native samples. "Synthetic" = simulated grid [14]. Events are inclusive calendar dates.

Series	Source and access	Variables available → used	Rows	Label source
WTI, Henry Hub	FRED (DCOILWTICO, DHHNGSP), direct CSV	one price → one price	about 1.5k business days each	Two documented market events, dated before modelling
French load, line flow	Gillioz et al., Zenodo 10.5281/zenodo.13378476	4,097 bus loads → 1 national sum; 8,375 lines → 28 border crossings → 1	43,656 / 26,184 hourly	Events built into the simulation
GB demand	NESO Data Portal (CKAN API), national demand	ND (MW) and others → ND only	140,240 half-hourly	Lockdown; December 2022 cold snap
ERCOT demand	EIA Hourly Grid Monitor, BA = ERCO, demand adjusted	demand → demand only	43,824 hourly	Texas Freeze; Winter Storm Elliott

Table 2: Dataset structure. "Variables available → used" states how many input channels exist and how many a detector sees. Raw data are re-downloadable from the sources; the repository ships acquisition scripts, not data.

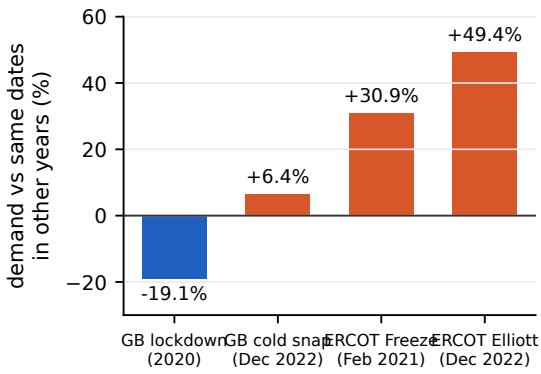


Figure 1: Change in mean demand over each labelled event relative to the same calendar dates in other years (values recorded in the data records; load shedding makes the Texas Freeze a lower bound).

labels are post hoc: for ERCOT, a period of all-time demand records between 27 June and 10 August 2023 [15] was identified *after* seeing detector output and is used only as a labelled sensitivity analysis, never in headline results. Labels differ greatly in how visibly they affect the series (Fig. 1). Mean demand over the event dates, compared with the same calendar dates in other years (as recorded when the data were acquired, and quoted as computed, not rounded or approximated), changed by −19.1% for the GB lockdown, +49.4% for ERCOT Elliott and +30.9% for the ERCOT Texas Freeze (recorded demand understates the Freeze because of forced load shedding), but only +6.4% for the GB cold snap, a weak label that we treat as such.

(A) Does the series deviate inside labelled events? For each event we compute the causal seasonal-naive residual. It is *weekly*, not monthly or quarterly: the baseline for a timestamp is the median of the same time of week in the previous four weeks (28 days, at least two of them available), the residual is the value minus that baseline, and it is divided by its robust spread (median absolute deviation, used only as a scale) and report the mean daily deviation inside the event, the share of its timestamps above $|z| > 3$, and the percentage of non-event days with a smaller deviation. This is independent of every detector. All four labelled events depart from the seasonal expectation (Table 3). The two ERCOT events and the GB cold snap are clear: their mean daily deviation lies above 95% of non-event days and 24–51% of their timestamps exceed $|z| > 3$, against 2–3% elsewhere. The GB lockdown is the weakest label: mean daily deviation 1.26, above 80% of non-event days, 9% of timestamps beyond $|z| > 3$. This is expected for a demand level that stayed low for ten weeks, because the four-week baseline adapts after the first weeks of the event, so the measure understates it; we report the number as computed and do not tune it.

(B) How incomplete are the labels? We count days outside every labelled event whose daily deviation is at least that of the weakest labelled event. These are *candidates* for missing labels, not claimed anomalies: holidays and weather can produce them. Reporting the count, rather than detector flags, quantifies label incompleteness without circularity. In GB, 546 of 2,842 days outside the events (19%) reach the weakest event's level (1.26), a lenient bar because the

Series	Event	Timestamps	Mean daily dev.	(non-event days)	Share $\lvert z\rvert > 3$	(non-event days)	% of non-event days below
GB	COVID	1,679	1.26	0.87	9%	2%	80%
GB	ColdSnap	240	2.66	0.87	24%	2%	98%
ERCOT	Texas Freeze	144	2.46	0.90	30%	3%	95%
ERCOT	Elliott spike	120	3.42	0.90	51%	3%	99%

Table 3: Analysis A: causal seasonal-naive residual inside each labelled event. "Mean daily dev." is the mean over the event's days of the daily mean $|z|$; the parenthesised columns give the same quantity for all days outside every labelled event of that series.

lockdown is a weak label; 61 of them lie between 24 December and 2 January, calendar effects that a four-week baseline does not model. In ERCOT, 69 of 1,816 days (3.8%) reach 2.46. Three of them (11–13 February 2021, daily deviation 2.7, 4.1 and 4.7) immediately precede the labelled start of the Texas Freeze on 14 February, which suggests that the labelled window starts late relative to the series' own departure; 31 January to 2 February 2023 (4.4, 4.3, 3.5) is a further strong, unlabelled departure. We have not checked these dates against operator records or weather data. The labels are therefore incomplete by this measure: the metrics treat every such day as normal, so alerts on them count as false positives (Fig. 2).

(C) Do rankings survive label errors? Detector scores are computed once; only the labels change: event starts or ends shifted by 1, 3 or 7 days (shifts at least as long as the event are skipped), both boundaries moved outward by the same amounts (dilation), events lengthened or shortened by half, and each event dropped in turn. We report whether the best detector and the full ranking (Kendall τ) are unchanged (Table 4); detectors use default configurations and three seeds, and the label sets are not independent, so the shares are descriptive.

GB. Boundary errors of up to seven days, dilation, shortening and lengthening leave the matrix profile best in all 18 such label sets (Kendall $\tau \geq 0.90$ for AUC-PR, ≥ 0.88 for timestamp F1). The one exception is dropping the lockdown, after which seasonal naive is best ($\tau = -0.24$ and -0.05). That label is 1,679 of the 1,919 labelled timestamps (87%); without it prevalence falls from 3.4% to 0.44% and every detector except seasonal naive scores AUC-PR between 0.003 and 0.004, close to prevalence. The GB ranking is therefore a statement about the lockdown label, not about boundary precision. **ERCOT.** Rankings are less stable. The best detector by AUC-PR changes in 2 of 17 label sets (dropping Winter Storm Elliott, and moving the Texas Freeze start three days later; both make the matrix profile best), with τ down to 0.62; by timestamp F1 it changes in 9 of 17 (47% unchanged), with τ down to 0.49. The top two, seasonal naive and matrix profile, swap under day-level changes, consistent with two events being too few to rank detectors (Sec. 8).

(D) Flags outside the labels: ERCOT summer 2023. Under the five-detector pipeline's clean-window protocol (B), the LSTM-AE flags 1,751 test timestamps, 95% of them in July–September 2023 and 38% inside the Tier 2 window (27 June to 10 August 2023, ERCOT's record-demand period); the Transformer is similar (93% and 37%). Their F1 against Tier 1 is exactly zero, and 0.43 (interval 0.32–0.53) and 0.42 once Tier 2 is labelled (Table 5). Under the forward split (A) the same models flag only 656 timestamps, all in July–September 2023 but almost none in Tier 2, and F1 stays near zero. An F1 of zero against Tier 1 therefore means that these detectors missed the two pre-specified events, *not* that they found nothing unusual: they respond to a period of record demand. We do not count this as a success, because Tier 2 was identified after seeing the output; it is reported as a sensitivity analysis, and the headline results use Tier 1 alone. The equal-budget runs (Sec. 8) are evaluated against Tier 1 only, so their ERCOT scores are lower bounds with respect to any additional true anomalies.

Limits. Documented external events are a coarser label than point annotations: the labelled window covers days in which many individual timestamps look normal, which is why timestamp-level recall is bounded and why per-event coverage is reported. Public benchmark labels have their own flaws [2]; Sec. 9 therefore treats them as noisy and reports sensitivity subsets.

5 Evaluation Protocol

5.1 Pipeline

(1) *Resolution.* A series is used at its native resolution unless stated; aggregation to a coarser resolution takes the mean over the block. (2) *Features.* Each timestamp has a four-feature vector (percentage return, rolling mean, rolling standard deviation and rolling skew of the return, window 30 days of time in the equal-budget runs and 30 rows in the published pipeline); the *residual* variant adds a causal seasonal-naive residual (above). Scalers are fitted on training rows only. (3) *Split.* Either a forward 70/30 split ("A") or a clean-window split ("B"): training on a stretch with no labelled event, testing on the rest. On real series nothing is injected into the test split; the synthetic designs inject events only into the test part and keep training clean. (4) *Fit and score.* The neural models are trained on the windows of the training rows (window 30 rows) and score a test row by the reconstruction error of the window ending there; classical detectors score rows (or, for the "window" variants,

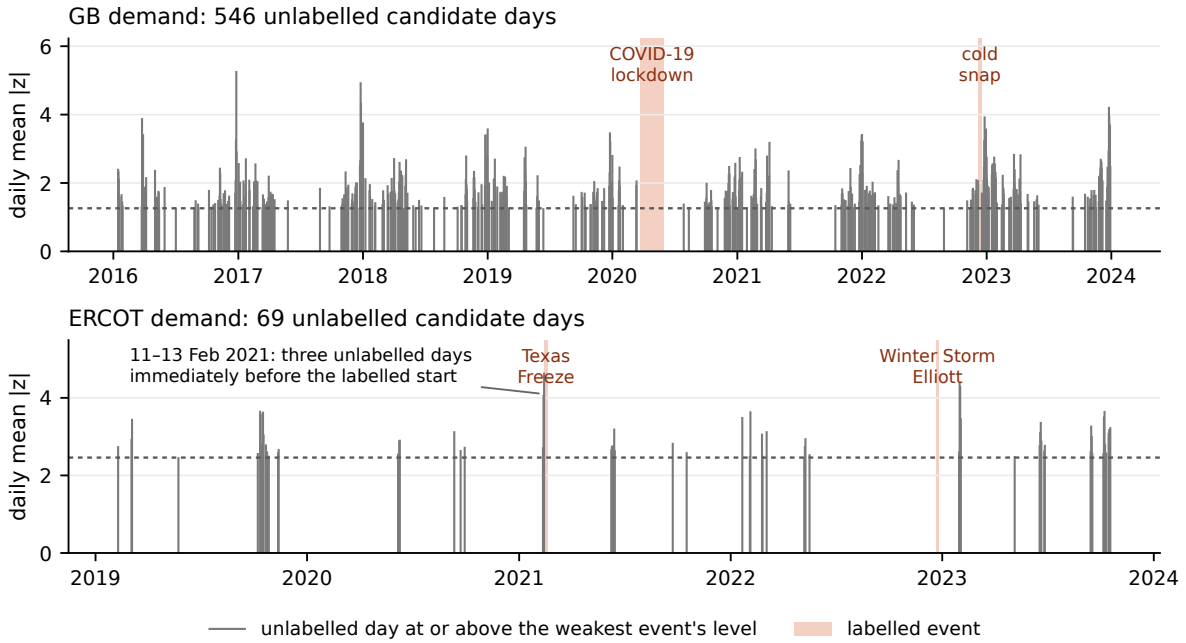


Figure 2: Unlabelled periods that the labels miss. Grey lines: days outside every labelled event (shaded) whose daily deviation from the seasonal-naive residual is at least that of the weakest labelled event (dotted line). They are candidates, not confirmed anomalies: GB shows year-end clusters (calendar effects) and ERCOT three days that immediately precede the labelled Texas Freeze.

Series	Metric	Baseline best	Best unchanged	Median τ	Min τ	Perturbations that change the best detector
GB	AUC-PR	matrix profile	95% of 19	1.00	-0.24	drop COVID (seasonal naive)
GB	timestamp F1	matrix profile	95% of 19	0.98	-0.05	drop COVID (seasonal naive)
ERCOT	AUC-PR	seasonal naive	88% of 17	0.81	0.62	drop Elliott spike (matrix profile); start later 3d (matrix profile)
ERCOT	timestamp F1	matrix profile	47% of 17	0.81	0.49	9 of 17; new best: seasonal naive

Table 4: Analysis C: stability of the detector ranking under label perturbations (Kendall τ of the full ranking against the unperturbed labels; 19 label sets for GB, 17 for ERCOT).

the same windows). All detectors in the equal-budget runs are *inductive*: every reference quantity comes from training rows. (5) *Alerts.* A fixed alert budget flags the top 5% of test scores. This uses the test score distribution (a quantile of the test scores) and is therefore an *alert-budget* evaluation, not a deployable threshold; DBSCAN as published flags its noise set and has no budget (its alert rate floats; Sec. 6). (6) *Labels to metrics.* Each timestamp inherits the label of the event window that contains it. All detectors are evaluated on the *same* rows (the intersection of the rows every model can score).

5.2 Detectors and settings

Table 6 lists the settings of the five original detectors. The equal-budget runs add, with the same 5% budget and the same features: a seasonal-naive residual score (|residual|), the one-line first-difference z-score of Wu and Keogh [2], PCA, k-NN and isolation forest on the same 30-step windows the neural models see, the matrix profile [20] (z-normalised nearest-neighbour distance to the clean subsequences, computed here with a matrix product, and checked against a brute-force reference in the tests), and the spectral residual [21]. Each family is given the default configuration plus 8 random configurations; selection uses a validation split of the clean training window into which synthetic events (level shifts, variance bursts, ramps and pulses of several durations) are injected, so real test events never influence a choice; a family is only tuned if validation can separate its configurations, otherwise its default is kept.

5.3 Metrics

Timestamp-level precision, recall and F1 count timestamps (the usual point-wise convention, cf. [1]). **Event-level** metrics count events and alerts: an alert is a maximal run of flags, an event is detected if any flag falls within its window widened by a tolerance (one day in the equal-budget runs), and precision is the fraction of alerts that overlap an event window, in the spirit of sequence-level scoring [10, 11]. The two are never combined: a mixed F1 built from timestamp-level precision and event-level recall is not an F1 of anything. **Per-event coverage** is the share of an event's timestamps that carry a flag. **AUC-PR** (average precision; chance level = prevalence π) is threshold-free, and we report **VUS-PR** point values from our own re-implementation (cross-checking against the official package is listed in Sec. 11). VUS-

Protocol	Detector	Flags	Flags in Jul–Sep 2023	Flags in Tier 2	F1, Tier 1	F1, Tier 1+2 [95% interval]
A	LSTM-AE	656	100%	1%	0.000	0.006 [0.000, 0.016]
A	Transformer	656	100%	0%	0.000	0.000 [0.000, 0.000]
A	IForest	657	70%	13%	0.000	0.089 [0.053, 0.125]
A	DBSCAN	582	1%	1%	0.097	0.046 [0.011, 0.098]
A	Rolling z	657	26%	10%	0.010	0.075 [0.054, 0.093]
B	LSTM-AE	1,751	95%	38%	0.000	0.426 [0.324, 0.528]
B	Transformer	1,751	93%	37%	0.000	0.417 [0.320, 0.521]
B	IForest	1,752	19%	4%	0.000	0.045 [0.025, 0.067]
B	DBSCAN	1,259	0%	0%	0.047	0.030 [0.008, 0.061]
B	Rolling z	1,752	6%	2%	0.002	0.028 [0.017, 0.039]

Table 5: ERCOT label tiers, five-detector pipeline (single run): where each detector's flags fall and F1 with Tier 1 only versus Tier 1 plus the post-hoc Tier 2 window. Protocol A: forward split; B: clean-window split. Intervals are day-block bootstrap.

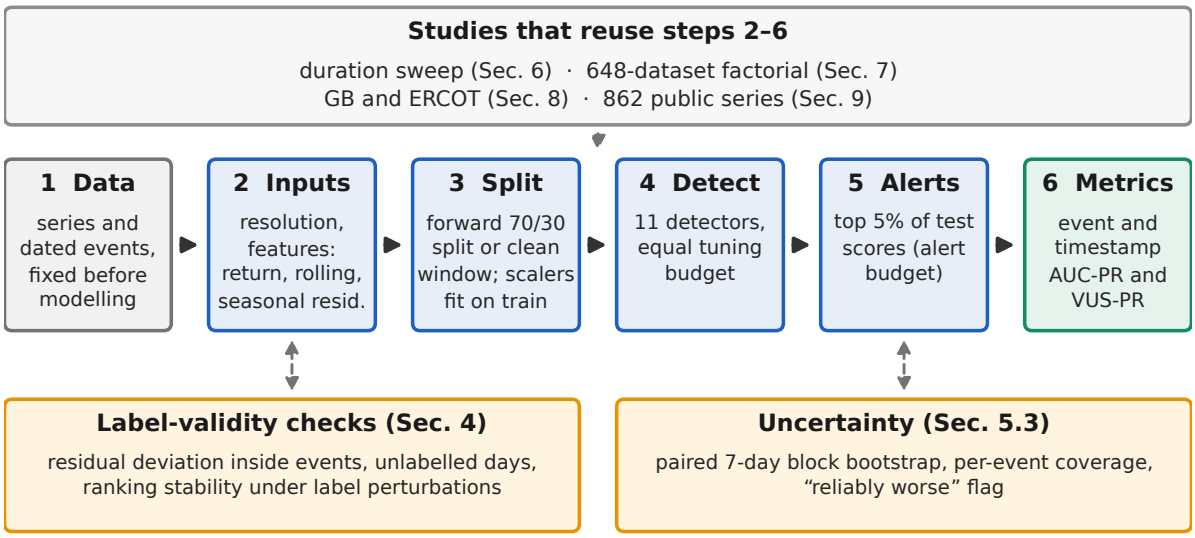


Figure 3: End-to-end evaluation pipeline. Steps 1–6 are the protocol of Sec. 5; the label-validity checks and the uncertainty analysis act on the same scores and labels, and every study in Secs. 6–9 reuses steps 2–6.

Detector	Settings and how flags are produced
LSTM-AE	hidden 32, latent 8; Adam 10^{-3}; 30 epochs; batch 32; window 30 rows. Flag = top 5% of test reconstruction error.
Transformer	d_{model} 32, 4 heads, 2 layers, FF 64; AdamW 5×10^{-4}, cosine; 30 epochs; batch 32; window 30. Flag = top 5% of test error.
IForest	sklearn, 100 trees, fitted on training rows. Flag = top 5% of test scores.
DBSCAN	ϵ = 90th percentile of 5-NN distance on train (Study 1: fixed ϵ=1.5); min_samples 5; **fitted on the test set** (transductive); flag = noise points, so the alert rate is *not* fixed.
"Rolling z"	Study 1: $\lvert z\rvert > 3$ with a rolling baseline. Studies 2–4: |return| above its 95th percentile on test (a fixed 5% budget on a one-feature score, *not* a rolling z as previously described).

Table 6: Settings of the five original detectors (from the study scripts). The equal-budget runs use the same neural settings; the sweep (Sec. 6) trains them for 20 epochs with batch 128.

PR [6] addresses a real weakness of timestamp-level AUC-PR: a score that peaks a few samples before or after a labelled boundary is counted as wrong. It replaces the binary label by one with a soft buffer of ℓ samples on each side of every event and averages the area under the precision–recall curve over buffer lengths $\ell = 0, \ldots, L$, a "volume" rather than an area. TSB-AD [7] recommends it as the most reliable measure. We do not claim that anything is wrong with the measure; the open points are ours: it adds a parameter (L, tied in the reference implementation to a period or window length), our implementation has not yet been compared with the official one, and the resampling needed for intervals is not the plain block bootstrap, so we report VUS-PR as point values only. Because a fixed budget $b = 5\%$ caps timestamp F1 at $F_1^{\max} = 2\min(\pi, b)/(\pi + b)$ (0.75 at $\pi = 3\%$), we report π and $F_1^{\max}$ with every result. **Uncertainty:** seeds do not capture the uncertainty from having few events, so we use a paired block bootstrap [12, 13] (7-day blocks, the same resampled rows for every detector, 200 replicates), report Δ to the best detector, and mark a detector "reliably worse" only if the interval of the difference excludes zero. The best detector is chosen post hoc, so such statements are exploratory. For range metrics a plain block bootstrap is invalid (it breaks events and lets buffers cross seams); we use event-preserving resampling units (Sec. 11).

6 Controlled Sweep over Duration and Resolution

Design. We generate 9.6 years of 30-min data (daily, weekly, annual seasonality; AR(1) residual; t_6 noise), clean for the first 60% (train) and with injected events in the final 40% (test). Event *duration* takes 2 h, 12 h, 1 d, 3 d, 7 d, 14 d. The number of events is chosen so that anomalous prevalence is constant at $\pi \approx 3\%$ (realised range 0.0299–0.0317), with at least three events in every cell. Events are either *level shifts* (+3 standard deviations) or zero-mean *variance bursts* (3 s.d.), start on day boundaries, and never overlap. The *same* series is analysed at 30 min, 1 h and 1 d (mean aggregation; cells where an event is shorter than one sample are omitted). Ten seeds vary noise and event placement. Detectors: Isolation Forest, DBSCAN as published, an inductive kNN-to-train distance, the |return| top-5% score (what Studies 2–4 call rolling z), a true rolling $z > 3$, a PCA window-reconstruction model (window 30 samples), and the LSTM autoencoder and Transformer autoencoder of Table 6 (window 30 samples, trained on the clean training rows, top-5% budget like the other budgeted detectors). To keep the sweep tractable the neural models are trained for 20 epochs with batch size 128 instead of 30 epochs and batch 32, and one model per seed and resolution is re-used for every event duration, because the training rows are identical across durations (events are injected into the test part only). Rolling features use a window fixed at 7 days of time.

What Results 1–4 are. They are four analyses of this one sweep, not four studies: the same 10 seeds × 16 duration–resolution cells per event type (6 durations at 30 min and 1 h, 4 at 1 d) × 8 detectors, with the same prevalence and alert budget. They differ only in the question asked of it: which detector is best (Result 1), how DBSCAN depends on resolution and duration (Result 2), what DBSCAN's alert rate does (Result 3), and whether thresholded F1 and threshold-free AUC-PR agree (Result 4). Studies 1–4 elsewhere in the paper are the real and simulated datasets of Sec. 3.

Result 1: the best detector depends on duration relative to the detector's time scale. For level shifts (Table 7, top), the rolling z is clearly best up to 1 d (AUC-PR 0.89, 0.97, 0.71) consistent with its 7-day trailing baseline not yet having adapted to the shift; from 3 d it collapses (0.33 at 3 d, 0.10 at 7 d), consistent with the baseline absorbing the event, and Isolation Forest is best (0.59, 0.78; a tie with kNN at 14 d). A window-30 PCA model peaks at 12 h–1 d (0.38, 0.34), consistent with the event filling its 15-hour window (mechanisms here are inferred, not separately tested). For variance bursts the picture is different: a per-point |return| rule is best up to 3 d, and windowed or learned models catch up (statistical ties) at 7–14 d. The ordering therefore changes with event type as well as duration, which the cross-dataset comparison could not separate. The two neural autoencoders are never best or second best at 30 min. For level shifts they rank third to eighth of eight, low up to 1 d (AUC-PR 0.04–0.10; chance is 0.03, except 0.17 for the Transformer at 12 h), rise to 0.53–0.58 at 7–14 d, and tie statistically with the best detector only at 14 d (0.57 and 0.58 against 0.64). For variance bursts they rank fourth to eighth, from 0.06–0.08 at 2 h to 0.92 at 14 d, behind the |return| rule and window PCA (0.99 at 14 d). The two models track each other closely except for 12 h shifts, and at daily sampling both stay low (mean AUC-PR over durations 0.09 for shifts, 0.05–0.07 for bursts). Their settings were reduced for the sweep (above), so this shows that they do not outperform simpler detectors under this protocol, not that no tuned neural model could.

Result 2: resolution alone does not reproduce the claimed DBSCAN decline. At fixed duration, DBSCAN's F1 changes by at most 0.021 (shifts) and 0.022 (bursts) between 30 min and 1 h, a doubling of sample count. Against duration, DBSCAN's F1 at 30 min is U-shaped, not monotone (shift: 0.35, 0.24, 0.26, 0.32, 0.35, 0.32 from 2 h to 14 d), and its AUC-PR is nearly flat (0.14–0.20). The proposed claim that DBSCAN falls monotonically with event sample count is not reproduced when prevalence, series and domain are held fixed.

Result 3: DBSCAN's alert rate floats, and for level shifts it dominates its F1. DBSCAN's realised alert rate is 11–24% for shifts and 9–21% for bursts, against 5% for the other detectors (Fig. 5). Across runs, its F1 correlates −0.92 with that alert rate for level shifts but +0.10 for bursts, so the artefact is event-type dependent. Cross-dataset DBSCAN F1 values from a transductive, floating-budget detector should not be read as detection quality.

Result 4: thresholded F1 and AUC-PR can disagree completely. The true rolling $z > 3$ flags at most 2.0% of points in any cell (mean over seeds), and at daily resolution its F1 is exactly 0 in every cell, while its AUC-PR is 0.97 at the 1-day event (Fig. 5). A fixed threshold, not the score, determined the F1.

7 Attribution on a Fully Crossed Semi-Synthetic Design

Real series cannot separate event type, duration, magnitude, resolution, background and feature set, because each real series brings one value of each. We therefore cross them. **Design** (pre-registered before the run): 3 backgrounds (strongly seasonal; weakly seasonal and noisy; strongly seasonal with trend) × 3 event types (level shift, variance burst, ramp) × 3 durations (4, 24, 120 h) × 3 magnitudes (3, 6, 12 residual standard deviations at the peak) × 2 resolutions

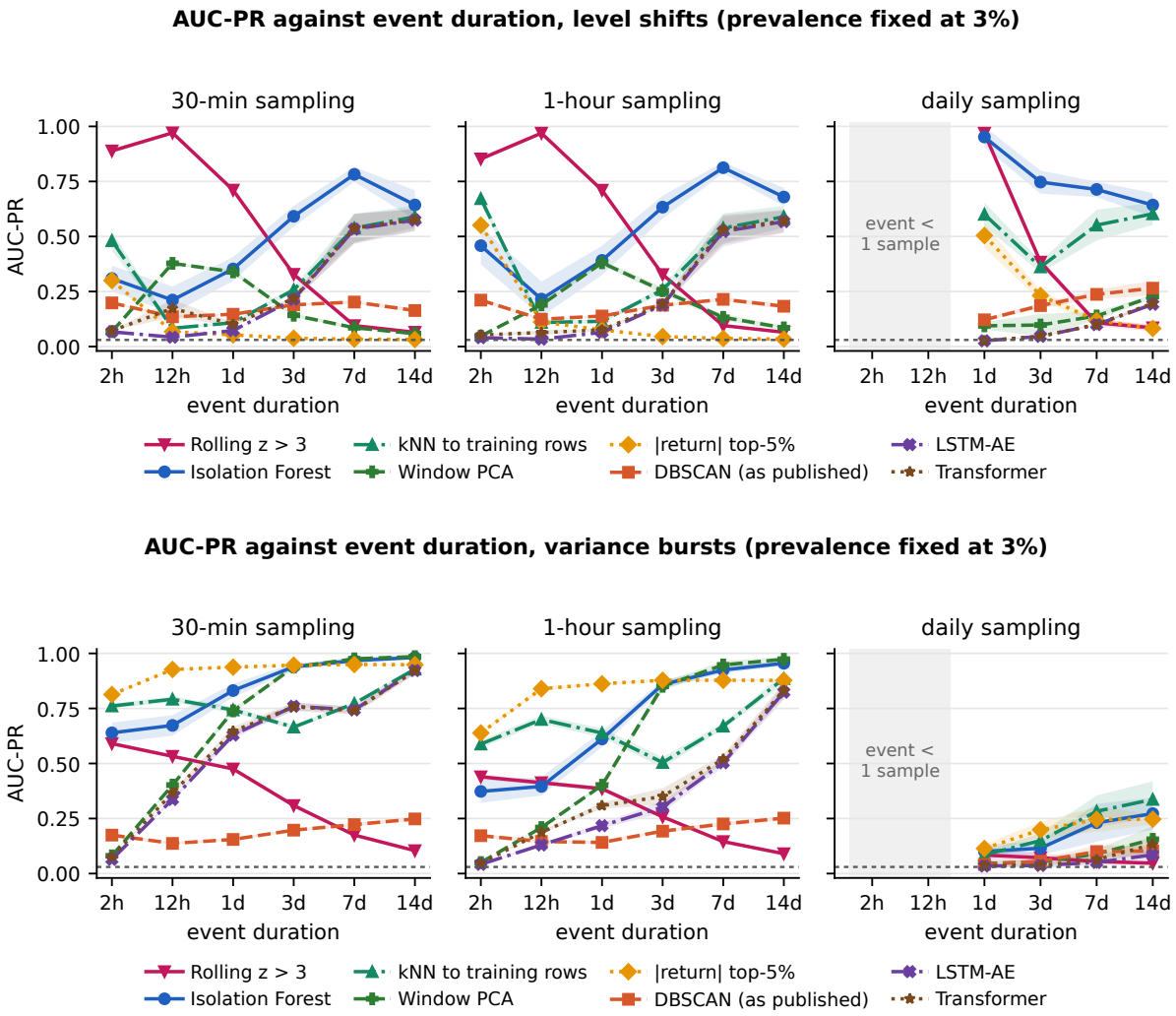


Figure 4: Threshold-free AUC-PR versus event duration at constant prevalence (dotted line = chance = π), for level shifts (top) and variance bursts (bottom), by sampling resolution. Bands: 95% t-intervals over 10 seeds; grey areas mark durations shorter than one sample.

(1 h, 4 h) × 4 replicates = 648 datasets, each with 120 clean training days and 90 test days containing three events. Nine detectors with default configurations, under both feature sets, are scored on identical rows (nine detectors × two feature sets = 11,664 results). The outcome is normalised AUC-PR (skill) = $(\mathrm{AP}-\pi)/(1-\pi)$, because lift is bounded by $1/\pi$ and π varies with duration. A sum-of-squares decomposition of the balanced design gives each factor's share of the variance (Table 8); detectors that do not read the feature columns return identical scores under both feature sets (a structural zero).

Result. The detector main effect explains only 19.9% of the variance; everything that involves the detector (main effect plus its two-way interactions) explains 47.4%, so more than half of detector-related variance is conditional on the setting. The interactions with duration, event type, magnitude and resolution sum to about 26%, more than the main effect itself. Under the rule fixed in advance this means that statements of the form "A beats B" are not portable across settings and must be stated per setting. Rankings are only moderately stable: the detector with the highest mean skill (a seasonal-naive residual) is best in 56% of 324 conditions, and eight of nine detectors win at least one. Event type is decisive: the seasonal-naive residual wins 76% of shift and 81% of ramp conditions but only 11% of burst conditions, where the first-difference one-liner wins 44%.

Caveat. The synthetic backgrounds are exactly seasonal with additive noise, the situation for which a seasonal-naive residual is built; its global win says little about real grids, and the shares describe this design and its level choices, not energy data in general. The feature-set effect is small here (0.5%) because only four of nine detectors read the features; on real data it was larger (Sec. 8).

Duration	Best (AUC-PR)	AUC-PR	2nd best	AUC-PR	DBSCAN AUC-PR	DBSCAN alert rate
2h	Rolling $z > 3$	0.89±0.01	kNN-to-train	0.48±0.04	0.20	12%
12h	Rolling $z > 3$	0.97±0.01	PCA window recon.	0.38±0.00	0.14	21%
1d	Rolling $z > 3$	0.71±0.01	IForest	0.35±0.05	0.15	19%
3d	IForest	0.59±0.05	Rolling $z > 3$	0.33±0.01	0.19	16%
7d	IForest	0.78±0.03	kNN-to-train	0.54±0.07	0.20	13%
14d†	IForest	0.64±0.07	kNN-to-train	0.59±0.05	0.16	11%

Duration	Best (AUC-PR)	AUC-PR	2nd best	AUC-PR	DBSCAN AUC-PR	DBSCAN alert rate
2h	|return| top-5%	0.81±0.01	kNN-to-train	0.76±0.01	0.17	16%
12h	|return| top-5%	0.93±0.00	kNN-to-train	0.79±0.01	0.14	21%
1d	|return| top-5%	0.94±0.00	IForest	0.83±0.03	0.15	19%
3d†	|return| top-5%	0.95±0.00	IForest	0.94±0.01	0.20	15%
7d†	PCA window recon.	0.98±0.01	IForest	0.97±0.01	0.22	13%
14d†	PCA window recon.	0.99±0.01	IForest	0.98±0.01	0.25	12%

Table 7: Best and second-best of the eight detectors by AUC-PR at 30 min (mean±95% interval over 10 seeds), with DBSCAN's AUC-PR and realised alert rate; the LSTM-AE and Transformer are never in the top two. Top: level shifts; bottom: variance bursts. † the two intervals overlap (a tie, not a win).

Source	% of variance in skill
Detector (main effect)	19.9
Event type	10.4
Duration × detector	10.1
Event type × detector	9.6
Event type × duration	6.3
Magnitude	4.5
Magnitude × detector	4.1
Resolution × detector	2.3
Feature set × detector	0.5
Background × detector	1.0
Remainder (higher-order terms, replicate noise)	24.0

Table 8: Share of the variance of detector skill (648 datasets, 9 detectors, 2 feature sets). Terms not shown are below 1%. Bootstrap intervals over the four replicates are within about 1.5 percentage points for every listed term.

8 Real Data: Baselines, Equal Budgets and Intervals

Setup. GB (half-hourly demand aggregated to hourly; events: 2020 lockdown, December 2022 cold snap; $\pi = 0.034$) and ERCOT (hourly; events: Texas Freeze, Winter Storm Elliott; $\pi = 0.0075$). Training windows are clean (GB 2018-06-01 to 2020-02-01; ERCOT 2021-06-01 to 2022-06-01). All detectors are inductive, use default configurations after the equal-budget validation described in Sec. 5 (tuning changed little and is omitted from the tables), five seeds, the same rows and the same 5% budget. DBSCAN as published is transductive with a floating alert rate and cannot be compared at an equal budget; its inductive analogue here is the k-NN distance to training rows. Tables 9 and 10 give AUC-PR and timestamp F1 under the published four-feature recipe and with the added seasonal residual.

Findings. (1) *Simple baselines match deep models.* On GB the matrix profile has the highest F1 (0.33, interval 0.18–0.43) and AUC-PR (0.21); with the seasonal residual the LSTM-AE reaches F1 0.23 (0.07–0.36), whose difference to the best has an interval that includes zero, while the Transformer is reliably worse. On ERCOT the seasonal-naive residual has the highest AUC-PR (0.24, 32 times prevalence) against 0.03 for the LSTM-AE and 0.02 for the Transformer. (2) *Feature design is first-order.* The matrix profile reads the raw values and gives the same result under both feature sets, which makes it a control: every feature-based detector changes. The LSTM-AE's GB F1 nearly doubles (0.125 to 0.233) and on ERCOT the LSTM-AE and Transformer go from F1 = 0 to nonzero, but window PCA *loses* AUC-PR on ERCOT (0.124 to 0.080), so the effect is not uniformly positive. (3) *Different detectors catch different events* (Table 11): on GB only the seasonal residual covers the weak December 2022 cold snap (0.65 of its timestamps) while the matrix profile and the neural models cover the lockdown but not the cold snap, consistent with level-blind or smoothed scores missing a small level change. (4) *Metrics disagree.* At the 5% budget on ERCOT, event-level F1 of the neural models (0.30 for the LSTM-AE, 0.44 for the Transformer) is higher than that of any other detector while their AUC-PR (0.03, 0.02) is low, so a ranking quoted without its metric is not informative. (5) *Tuning did not help.* For the neural families validation could not separate configurations (the default was kept), and for the matrix profile the selected window lowered GB F1 from 0.33 to 0.14: selection on synthetic validation events does not reliably predict ranking on real events.

Five-detector pipeline, one metric. Table 12 reports the timestamp-level F1 of the five-detector pipeline from the stored result files, one metric throughout (single run per dataset, seed 42, no intervals: descriptive only). On WTI the best detector under the consistent metric is the LSTM autoencoder (0.590), not DBSCAN; DBSCAN's F1 is 0.344, whereas a mixed timestamp-precision/event-recall score gives 0.695. The selection rule and the "detection gap" are **not supported**: the gap's stated range contains GB lockdown (LSTM F1 0.282 pooled over two events) and French line flow (0.408), and the rule was not tested on held-out events before (Sec. 9).

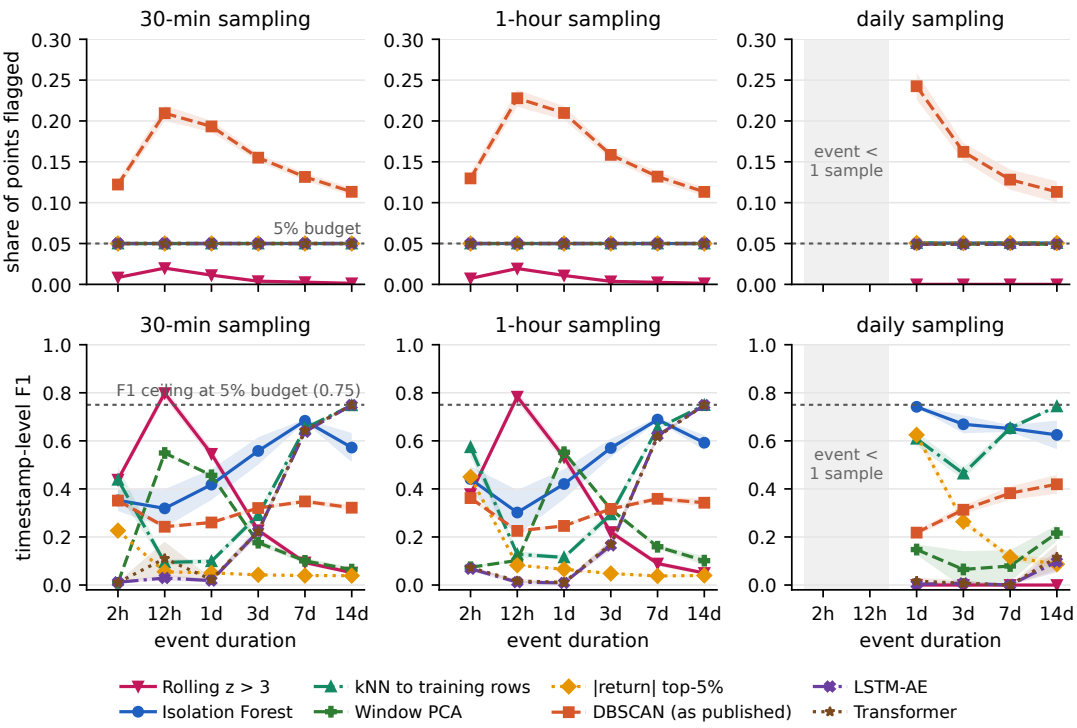


Figure 5: Level shifts. Top: realised alert rate; the 5% line is the budget used by all other detectors, and DBSCAN (as published) exceeds it in every cell. Bottom: timestamp-level F1 under the published rule; the dotted line is the F1 ceiling at a 5% budget and $\pi = 3\%$ (0.75).

9 Held-Out Test of the Rule on Public Series

The rule and the test. The rule, as proposed in the five-detector study, says that reconstruction models gain over density models as an event spans more native samples (so the best detector should follow event duration relative to sampling resolution). We tested it on the univariate TSB-AD series [7]: for each series, the training prefix named in the file name, a period estimated by autocorrelation, nine detectors scored on the same test rows, and AUC-PR per detector. Window PCA (reconstruction) and k-NN-to-train (density) stand in for the two model classes, because the neural models are too slow for 870 series. The primary statistic is the partial Spearman correlation between log median event length (samples) and the AUC-PR difference, controlling for log prevalence, computed within each source dataset, combined by Fisher z over datasets with at least eight series, with a bootstrap over datasets. The decision rule (supported / contradicted / not supported by the 95% interval) was committed before any benchmark series was analysed.

Result. 862 series from 23 source datasets (17 with at least eight series; 8 of 870 files excluded: 7 with no anomaly in the test part, 1 detector failure): $\rho = -0.047$, 95% interval $[-0.163, 0.139]$, which contains zero, so the rule is **not supported**. Sensitivity subsets give the same answer (Table 13). Per-dataset estimates disagree in sign. Window PCA beats k-NN on only 25% of series (median AUC-PR difference -0.094), so the stand-in pair is a weak proxy for the original comparison; the test therefore says that event length does not predict this advantage, not that no reconstruction model could show one. A first run that omitted 40 series (a flat training prefix broke our period estimator) gave $\rho = -0.060$ with the same verdict.

10 Discussion: a Checklist

The results suggest that an energy anomaly benchmark should report: (1) event-level and timestamp-level metrics separately; (2) prevalence, realised alert rate and $F_1^{\max}$; (3) a threshold-free metric next to any fixed-budget F1; (4) inductive fitting, or explicit disclosure of transductive fits; (5) simple baselines (a seasonal-naive residual, a one-line difference score) and the input features given to every model; (6) per-event results and intervals that account for the number of events, with no ranking from one or two events; (7) the label provenance, and tests of label validity that do not rely on detectors; (8) independent variation of duration, resolution and event type when claiming a duration effect, and constant prevalence when varying duration.

Detector	4-feature recipe AUC-PR [95%]	4-feature recipe F1 [95%]	recipe + seasonal residual AUC-PR	recipe + seasonal residual F1 [95%]
Matrix profile	0.213 [0.077, 0.344]	0.332 [0.184, 0.432]	0.213	0.331 [0.192, 0.436]
Seasonal-naive resid.	–	–	0.084	0.176 [0.063, 0.278]
LSTM-AE	0.112 [0.030, 0.271]	0.125 [0.000, 0.287]*	0.149	0.233 [0.065, 0.361]
Transformer	0.061 [0.026, 0.136]	0.095 [0.001, 0.208]*	0.080	0.124 [0.009, 0.251]*
Window PCA	0.026 [0.013, 0.041]	0.001 [0.000, 0.002]*	0.050	0.106 [0.016, 0.182]*
Isolation Forest	0.054 [0.022, 0.095]	0.059 [0.017, 0.101]*	0.090	0.144 [0.052, 0.214]*
Window IForest	0.069 [0.027, 0.145]	0.080 [0.001, 0.186]*	0.091	0.125 [0.030, 0.202]*
kNN-to-train	0.054 [0.024, 0.096]	0.002 [0.000, 0.007]*	0.059	0.020 [0.007, 0.035]*
Window kNN	0.058 [0.025, 0.101]	0.003 [0.000, 0.009]*	0.071	0.015 [0.001, 0.031]*
$\lvert\Delta x\rvert$ one-liner	0.032 [0.015, 0.049]	0.020 [0.009, 0.030]*	0.032	0.020 [0.009, 0.029]*
Rolling z	0.035 [0.017, 0.054]	0.027 [0.016, 0.036]*	0.035	0.027 [0.015, 0.037]*
Spectral residual	0.034 [0.017, 0.054]	0.029 [0.017, 0.039]*	0.034	0.029 [0.017, 0.037]*

Table 9: GB demand ($\pi = 0.034$). AUC-PR and timestamp F1 at a 5% budget, mean over five seeds; brackets are paired 7-day-block bootstrap intervals. * the interval of the difference to the best detector in that column excludes zero (exploratory: the best is chosen post hoc). AUC-PR intervals are not available for the residual feature set because that run predates the interval code.

Detector	4-feature recipe AUC-PR [95%]	4-feature recipe F1 [95%]	recipe + seasonal residual AUC-PR	recipe + seasonal residual F1 [95%]
Matrix profile	0.131 [0.026, 0.288]	0.132 [0.000, 0.263]	0.133	0.134 [0.000, 0.257]
Seasonal-naive resid.	–	–	0.239	0.125 [0.000, 0.254]
LSTM-AE	0.004 [0.001, 0.010]	0.000 [0.000, 0.000]	0.027	0.097 [0.000, 0.237]
Transformer	0.008 [0.002, 0.021]	0.000 [0.000, 0.000]	0.019	0.052 [0.000, 0.145]
Window PCA	0.124 [0.002, 0.299]	0.055 [0.000, 0.116]	0.080	0.060 [0.000, 0.171]
Isolation Forest	0.005 [0.001, 0.013]	0.000 [0.000, 0.001]	0.013	0.034 [0.000, 0.104]
Window IForest	0.006 [0.001, 0.014]	0.000 [0.000, 0.000]	0.032	0.044 [0.000, 0.129]
kNN-to-train	0.006 [0.002, 0.016]	0.001 [0.000, 0.003]	0.014	0.032 [0.000, 0.090]
Window kNN	0.008 [0.002, 0.020]	0.000 [0.000, 0.000]	0.022	0.065 [0.000, 0.161]
$\lvert\Delta x\rvert$ one-liner	0.009 [0.001, 0.017]	0.002 [0.000, 0.004]	0.009	0.002 [0.000, 0.005]
Rolling z	0.011 [0.003, 0.021]	0.024 [0.000, 0.049]	0.012	0.024 [0.000, 0.052]
Spectral residual	0.008 [0.002, 0.019]	0.012 [0.000, 0.024]	0.008	0.017 [0.000, 0.034]

Table 10: ERCOT demand ($\pi = 0.0075$); same conventions as Table 9. No detector is reliably worse than the best: two events cannot rank detectors.

11 Limitations

- Two events per real series cannot rank detectors; the real-data comparison is evidence about *which things matter*, not about a winner. Documented operator-declared event lists, not used here, could raise the count.
- All detectors are univariate; relations between series (the 4,097 grid buses, other balancing areas) are not used.
- Fixed-budget flags use a quantile of the test scores; this is an alert-budget evaluation, not a deployable threshold.
- The factorial and sweep designs are synthetic; their backgrounds favour seasonal-naive scores; level choices were taken from the real-data runs.
- The neural models are in the duration sweep (with reduced training, Sec. 6) but not in the factorial or public-benchmark tests (cost), so the public test uses window PCA and k-NN as stand-ins. Stronger recent detectors (foundation models, Series2Graph, NORMA, TranAD) are not evaluated.
- VUS-PR is our re-implementation; agreement with the official package is not yet measured **[TO RUN: run `experiments/crosscheck_official.py`]**. VUS-PR bootstrap intervals from the runs of 2 Nov 2026 used an invalid resampling and are not reported; the fixed procedure exists but the runs have not been repeated **[TO RUN: repeat the tuning runs and report VUS-PR intervals]**.
- The published neural numbers of the five-detector pipeline (Table 12) build windows across the gap of the clean-window split (about 30 windows) and were not regenerated; the equal-budget runs use segment-aware windows.
- Label validity (Sec. 4) is assessed on four events in two series. Analysis A and B use a seasonal-naive residual, so they are blind to anomalies that the baseline absorbs; candidate days in B were not checked against operator records; the perturbations in C are a stress test, not a model of how real annotation errors arise.
- DBSCAN is not in the equal-budget runs (transductive, no budget); the sweep (Sec. 6) covers it.

12 Conclusion

A duration–resolution rule that looked well supported by four datasets was the product of mixed metrics, floating alert budgets and variables that moved together; it fails a controlled sweep and a held-out test on 862 public series. What remains is more modest and more useful: detector rankings on energy data are conditional on event type, duration, magnitude and input features, simple baselines are competitive with deep autoencoders, and a handful of events cannot establish a winner. The protocol, the label-validity analyses and the released tests are meant to let others measure those dependences on their own series rather than assume them.

Detector	COVID	ColdSnap	Events hit
Matrix profile	0.47	0.00	1/2
Seasonal-naive resid.	0.15	0.65	2/2
LSTM-AE	0.33	0.00	1/2
Transformer	0.17	0.00	1/2
Window PCA	0.15	0.00	1/2
Isolation Forest	0.20	0.01	2/2
Window IForest	0.18	0.00	1/2
kNN-to-train	0.03	0.00	1/2
Window kNN	0.02	0.00	1/2
$\lvert\Delta x\rvert$ one-liner	0.02	0.05	2/2
Rolling z	0.03	0.05	2/2
Spectral residual	0.03	0.04	2/2

Detector	Texas Freeze	Elliott spike	Events hit
Matrix profile	0.68	0.30	2/2
Seasonal-naive resid.	0.38	0.59	2/2
LSTM-AE	0.37	0.36	2/2
Transformer	0.26	0.13	2/2
Window PCA	0.37	0.06	2/2
Isolation Forest	0.23	0.01	2/2
Window IForest	0.30	0.01	2/2
kNN-to-train	0.09	0.16	2/2
Window kNN	0.24	0.25	2/2
$\lvert\Delta x\rvert$ one-liner	0.01	0.01	2/2
Rolling z	0.07	0.12	2/2
Spectral residual	0.06	0.07	2/2

Table 11: Per-event coverage (share of an event's timestamps flagged; residual feature set, mean over seeds). "Events hit" counts events with any flag, so read it with the coverage values: a detector can hit an event with coverage near zero.

Data	Split	LSTM	Trf.	IF	DBSCAN	z
WTI	forward	**0.590**	0.551	0.443	0.344	0.055
Henry Hub	forward	0.000	0.000	0.127	**0.304**	0.068
FR Load	A	0.060	0.092	0.029	**0.222**	0.055
FR Load	B	0.101	0.051	0.077	**0.196**	0.075
FR Line Flow	B	**0.408**	**0.408**	0.103	0.148	0.081
GB demand	A	0.000	0.000	0.000	**0.039**	0.019
GB demand	B	**0.282**	0.142	0.111	0.068	0.027
ERCOT demand	A	0.000	0.000	0.000	**0.097**	0.010
ERCOT demand	B	0.000	0.000	0.000	**0.047**	0.002

Table 12: Timestamp-level F1 of the five-detector pipeline (best per row in bold); Split: forward 70/30 (A or "forward") or clean-window (B). Single run, seed 42.

Reproducibility

Code, tests (including regression tests for the event-end-day, window-seam and VUS-bootstrap bugs), result files, preregistrations (`docs/STEP8_PREREGISTRATION.md`, `docs/STEP9_PREREGISTRATION.md`, committed before the corresponding runs) the table generator (`paper/revised/make_tables.py`) and the figure generator (`paper/revised/make_figures.py`) are at https://github.com/suyounas/energy-anomaly-benchmark-4d [22]; release `v1.0.0` (commit `87b870b92b`), archived at https://doi.org/10.5281/zenodo.23981052. All tables are generated by `paper/revised/make_tables.py` from CSVs.

Acknowledgment

This work is motivated by the port-congestion work of Andrew Howe at Wood Mackenzie [23]. The idea of treating anomaly-detection results as an evaluation problem was further influenced by the author's participation in the ATLAS search that used unsupervised machine learning for anomaly detection [24]. That was collaborative work of the ATLAS Collaboration at CERN; this paper uses no ATLAS data or code and does not represent the Collaboration.

Subset	Series	Datasets	ρ	95% interval
All series (primary)	862	17	-0.047	[-0.163, 0.139]
Clean training prefix only	691	13	0.008	[-0.130, 0.188]
Event length > 3 samples	561	15	-0.064	[-0.207, 0.109]
Period found only	604	14	-0.049	[-0.176, 0.229]

Table 13: Primary test and exploratory sensitivity subsets (same statistic). "Clean training prefix" excludes series with anomalies inside the training prefix (171 of 862 have some).

References

[1] S. Kim, K. Choi, H.-S. Choi, B. Lee, S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proc. AAAI*, vol. 36, no. 7, pp. 7194–7201, 2022. https://ojs.aaai.org/index.php/AAAI/article/view/20680

[2] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 3, pp. 2421–2429, 2023. https://doi.org/10.1109/TKDE.2021.3112126

[3] S. Schmidl, P. Wenig, T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *PVLDB*, vol. 15, no. 9, pp. 1779–1797, 2022. https://doi.org/10.14778/3538598.3538602

[4] P. Wenig, S. Schmidl, T. Papenbrock, "TimeEval: a benchmarking toolkit for time series anomaly detection algorithms," *PVLDB*, vol. 15, no. 12, pp. 3678–3681, 2022. https://doi.org/10.14778/3554821.3554873

[5] J. Paparrizos, Y. Kang, P. Boniol, R. S. Tsay, T. Palpanas, M. J. Franklin, "TSB-UAD: an end-to-end benchmark suite for univariate time-series anomaly detection," *PVLDB*, vol. 15, no. 8, pp. 1697–1711, 2022. https://doi.org/10.14778/3529337.3529354

[6] J. Paparrizos, P. Boniol, T. Palpanas, R. S. Tsay, A. Elmore, M. J. Franklin, "Volume under the surface: a new accuracy evaluation measure for time-series anomaly detection," *PVLDB*, vol. 15, no. 11, pp. 2774–2787, 2022. https://doi.org/10.14778/3551793.3551830

[7] Q. Liu and J. Paparrizos, "The elephant in the room: towards a reliable time-series anomaly detection benchmark," in *NeurIPS Datasets and Benchmarks Track*, 2024. https://proceedings.neurips.cc/paper_files/paper/2024/file/c3f3c690b7a99fba16d0efd35cb83b2c-Paper-Datasets_and_Benchmarks_Track.pdf

[8] Z. Z. Darban, G. I. Webb, S. Pan, C. C. Aggarwal, M. Salehi, "Deep learning for time series anomaly detection: a survey," *ACM Comput. Surv.*, 2024. https://doi.org/10.1145/3691338

[9] A. Lavin and S. Ahmad, "Evaluating real-time anomaly detection algorithms — the Numenta Anomaly Benchmark," in *Proc. IEEE ICMLA*, pp. 38–44, 2015. https://doi.org/10.1109/ICMLA.2015.141

[10] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, T. Soderstrom, "Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding," in *Proc. ACM SIGKDD*, pp. 387–395, 2018. https://doi.org/10.1145/3219819.3219845

[11] N. Tatbul, T. J. Lee, S. Zdonik, M. Alam, J. Gottschlich, "Precision and recall for time series," in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018. https://proceedings.neurips.cc/paper/2018/hash/8f468c873a32bb0619eaeb2050ba45d1-Abstract.html

[12] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *Ann. Statist.*, vol. 17, no. 3, pp. 1217–1241, 1989. https://doi.org/10.1214/aos/1176347265

[13] D. N. Politis and J. P. Romano, "The stationary bootstrap," *J. Amer. Statist. Assoc.*, vol. 89, no. 428, pp. 1303–1313, 1994. https://doi.org/10.1080/01621459.1994.10476870

[14] M. Gillioz, G. Dubuis, P. Jacquod, "A large synthetic dataset for machine learning applications in power transmission grids," *Scientific Data*, vol. 12, art. 168, 2025. https://doi.org/10.1038/s41597-025-04479-x

[15] ERCOT, "2023 peak demand records" (records are unofficial until final settlement). https://www.ercot.com/static-assets/data/news/content/a-peak-demand/2023/all-time-records.htm

[16] F. T. Liu, K. M. Ting, Z.-H. Zhou, "Isolation forest," in *Proc. IEEE ICDM*, pp. 413–422, 2008. https://doi.org/10.1109/ICDM.2008.17

[17] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997. https://doi.org/10.1162/neco.1997.9.8.1735

[18] A. Vaswani et al., "Attention is all you need," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017. https://arxiv.org/abs/1706.03762

[19] M. Ester, H.-P. Kriegel, J. Sander, X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. ACM SIGKDD*, pp. 226–231, 1996. https://dl.acm.org/doi/10.5555/3001460.3001507

[20] C.-C. M. Yeh et al., "Matrix Profile I: all pairs similarity joins for time series," in *Proc. IEEE ICDM*, 2016. https://doi.org/10.1109/ICDM.2016.0179

[21] H. Ren et al., "Time-series anomaly detection service at Microsoft," in *Proc. ACM SIGKDD*, 2019. https://doi.org/10.1145/3292500.3330680

[22] S. Younas, "energy-anomaly-benchmark-4d" (code, tests, results). https://github.com/suyounas/energy-anomaly-benchmark-4d

[23] A. Howe, "Innovation & Analytics Lab data science (describing a port-constraint proof of concept)," Wood Mackenzie, Medium, 25 Jul. 2022. https://medium.com/woodmac/innovation-analytics-lab-data-science-8c2d79deca79

[24] ATLAS Collaboration, "Search for new phenomena in two-body invariant mass distributions using unsupervised machine learning for anomaly detection at $\sqrt{s} = 13$ TeV with the ATLAS detector," *Phys. Rev. Lett.*, vol. 132, 081801, 2024. https://arxiv.org/abs/2307.01612